

Fail Thee Well

WE HAVE SPENT a lot of time discussing errors, mistakes, and failures. Since failure is inevitable, we would all benefit from learning how to fail better.

On a trading desk, you are taught to expect to be wrong. Surprisingly, that attitude is rare elsewhere in finance. This is a shame because a healthy outlook on failure benefits businesses, governments, and just about everyone.

High stakes make aviation an excellent subject for the study of failure. In other fields, errors may be subtle, and the results may not be recognized for years. When there is a flying failure, planes fall out of the sky, and footage of the wreckage is on the evening news.^{[452](#)}

Matthew Syed points this out in *Black Box Thinking*.^{[453](#)} Aviation is an open, data-rich system, with statistics going back a century: In 1912, the US Army had 14 pilots, and even before the war, more than half (eight) would die in crashes. The Army (this was before there was a US Air Force) set up an aviation school to teach pilots how to fly more successfully. The school had an unfortunate 25% mortality rate.

Fast-forward a century. Syed observed that in 2013, there were 36.4 million commercial flights worldwide carrying 3 billion passengers. That year, there were only 210 commercial aviation fatalities. For some context, 1 million flights resulted in 0.41 accidents—an average of 2.4 million flights were needed for just one single accident. In 2017, there were *zero* commercial airline passenger deaths; none in 2022, 2023, or 2024 (up to the time of going to press). That is an astounding improvement over the course of a century.

How did the industry achieve this? By being self-critical and learning from every accident. The industry extensively studies each crash or near miss.